

Threat Model — STRIDE + LINDDUN

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Master technical specification — Volume 5 (Security & Privacy), nano-detail document **threat-model**. Deepens [04-security-privacy-pki.md] (the security spine: zero-trust, identity, PKI, no-logging, kill-switch, PQ) into an implementation-grade threat model: a STRIDE decomposition **per component**, a LINDDUN **privacy** decomposition, the attacker-class taxonomy, the trust-boundary diagram, every threat mapped to a concrete spec mechanism, an assets × threats × mitigation matrix, and an honest residual-risk register. SPEC ONLY — this document *describes* the security properties the build must have; it does not build the product. Sources are cited inline by id: [04_ARCH §N] (HelixVPN-Architecture-Refined.md), [04_P1 §N] (HelixVPN-Phase1-MVP.md), [04_P2 §N] (Phase-2 PQ/transport notes), [research-pki_pq_nat] (scratchpad/kb), [SYNTHESIS §N], and sibling specs by filename. Every invariant id S1–S11 is the one defined in [04-security-privacy-pki.md §0.1]; P1–P8 in [v03-control-plane/svc-policy.md §0.1]; I5/I6/R1–R4 in [v02-data-plane/routing-and-addressing.md §0.2]. Any claim not grounded in the evidence base is flagged **UNVERIFIED** per constitution §11.4.6 — never fabricated.

Table of contents

- 0. Position, ownership & what this document delivers
- 1. Methodology, scope & assumptions
- 2. Asset register — what we are protecting
- 3. Trust boundaries
- 4. Attacker classes
- 5. STRIDE per component
 - 5.1 Connector
 - 5.2 Gateway — control plane
 - 5.3 Gateway — Rust edge (data plane)
 - 5.4 Client device
 - 5.5 Coordinator
 - 5.6 PKI / tenant CA
 - 5.7 Data plane (the tunnel on the wire)
- 6. LINDDUN privacy threats
- 7. Threat → mitigation map (every threat to a spec mechanism)

- 8. Assets × threats × mitigation matrix
 - 9. Attack trees (two worked)
 - 10. Residual risks — honestly stated
 - 11. Test & validation mapping (§11.4.169)
 - Sources verified
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0. Position, ownership & what this document delivers

HelixVPN is a self-hostable overlay network with a privacy-VPN front end: three roles — **Connector** (outbound-only, advertises CIDs) $\rightleftharpoons$ **Gateway** (public VPS: control + data plane) $\rightleftharpoons$ **Client** (Mullvad-style end-user app) [04_ARCH §1.1, SYNTHESIS §1]. The security spine is owned by [04-security-privacy-pki.md]; **this document owns the threat analysis that justifies every mechanism in that spine**. A mechanism with no mapped threat is a candidate for removal; a threat with no mapped mechanism is a release blocker.

This document **owns**: the STRIDE/LINDDUN decompositions (§5/§6), the attacker-class model (§4), the trust-boundary diagram (§3), the asset register (§2), the threat→mitigation map (§7), the assets×threats×mitigation matrix (§8), and the residual-risk register (§10).

It does **not** own: the mechanisms themselves (the enrollment protocol, the PKI lifecycle, the kill-switch state machine, the PQ handshake — all [04-security-privacy-pki.md]); the policy compiler ([v03-control-plane/svc-policy.md]); the WG crypto core and transports (doc 01); the platform tunnel shims (doc 06). It *references* those by id and asserts what each defends against.

0.1 How a threat is recorded here (the discipline)

Each threat carries: a stable id (T-<COMPONENT>-<STRIDE-letter>-<n> or LP-<letter>-<n>), the STRIDE/LINDDUN category, the attacker class(es) that can reach it (§4), the asset(s) at risk (§2), the **mitigation** (a cited spec mechanism), and an honest **residual** note (§11.4.6 — "mitigated" never means "eliminated" unless proven). A mitigation that is only "planned/Phase-2" is labelled so; claiming an unshipped control as live is itself a §11.4 bluff at the threat-model layer.

1. Methodology, scope & assumptions

1.1 Two complementary frameworks

Framework	Question it answers	Applied in
STRIDE (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege)	<i>security</i> threats per component/data-flow crossing a trust boundary	§5

Framework	Question it answers	Applied in
LINDDUN (Linking, Identifying, Non-repudiation, Detecting, Data disclosure, Unawareness, Non-compliance)	<i>privacy</i> threats — the harms a privacy VPN exists to prevent	§6

STRIDE alone would miss the privacy core: a system can be STRIDE-clean (no spoofing, no tampering) and still **link** a user to the sites they visit. The no-logging invariant (S6) is a LINDDUN mitigation that STRIDE does not even ask about — hence both frameworks [research-pki_pq_nat].

1.2 Scope

In scope: every trust boundary in §3 — the hostile network, the client device, the gateway (control plane + edge + CA), and the connector. Out of scope (stated, not hidden): physical attacks on the operator's datacenter (deployment doc 09's concern), supply-chain compromise of upstream dependencies beyond the signed-image floor (S8, Phase-3 reproducible builds [04_ARCH §12]), and the security of the operator's chosen IdP (HelixVPN is an OIDC Relying Party, never an IdP — [04-security-privacy-pki.md §2.1]).

1.3 Assumptions (each is a load-bearing precondition — a false assumption is a finding)

#	Assumption	If false →
A1	The tenant CA private key is held in a KMS/HSM and never enters control-plane process memory in the recommended deployment (S11).	total identity compromise — §10 R-CA
A2	The device OS keystore (Keychain / Android Keystore / TPM / DPAPI) is not already compromised when keys are sealed (S2, [04-security-privacy-pki.md §3.3]).	device-key exfiltration — §10 R-DEV
A3	WireGuard's Curve25519 + ChaCha20-Poly1305 core is sound against a <i>classical</i> adversary today; quantum is addressed by hybrid PQ (S10).	data confidentiality — §10 R-PQ
A4	TLS 1.3 mutual auth with a pinned tenant CA is sound for the control channel (S4).	control-channel MITM — §5.2
A5	The operator runs the published rootless-Podman hardened images, not a hand-modified host (S8).	edge container escape — §5.3

2. Asset register — what we are protecting

Threats are scored against concrete assets. The asset's **sensitivity** drives mitigation priority (§11.4.132 risk-ordering applies to the security backlog too).

Asset id	Asset	Where it lives	Sensitivity	Why it matters
AS-CA	Tenant CA private key	KMS/HSM (S11)	Critical	Signs every device cert; its theft mints arbitrary trusted identities. The "one true secret" [04_ARCH §10].
AS-DB	Postgres control-plane truth (identity, policy, topology)	Gateway DB	Critical	The source of <i>who exists</i> and <i>who may reach what</i> ; corruption = policy bypass; exfil = membership graph disclosure.
AS-WGK	Per-device WG static private key	Device OS keystore (S2)	Critical	Decrypts that device's tunnel; never leaves the device by design.
AS-LEAF	Device leaf cert private key (Ed25519)	Device OS keystore	High	Authenticates the control channel; ≤24 h lifetime caps blast radius (S4).
AS-MAP	A device's policy-filtered NetworkMap	In transit + device memory	High	Reveals the peers/routes that device may reach (need-to-know, S3); leak = topology disclosure for that device's slice.
AS-TOKEN	Enroll token (plaintext, shown once)	Out-of-band to the user; only Argon2id hash stored	High	Whoever holds it can enroll a device into the tenant (§2.2).
AS-TRAFFIC	User traffic content + metadata (dst, timing, volume)	On the wire (ciphertext) + transiently in the edge	Critical (privacy)	The thing a privacy VPN exists to protect; never durably stored (S6).
AS-AUDIT	audit_events (control actions)	Postgres	Medium	Tamper or deletion hides operator/insider abuse (S7).
AS-PRESENCE		Redis (TTL'd)	Medium (privacy)	Coarse "who is online now";

Asset id	Asset	Where it lives	Sensitivity	Why it matters
	Ephemeral presence/session state			never a connection log; expires.
AS-PSK	PQ-derived session PSK	Device + edge memory, per-session	High	Adds harvest-now-decrypt-later resistance (S10); ephemeral, never persisted.

3. Trust boundaries

A trust boundary is any edge a data flow crosses between principals of different trust level. Each numbered boundary TB-n is where STRIDE is most productive — it is precisely where an attacker tries to spoof, tamper, or escalate.

```

flowchart TB
    subgraph Z0["Zone 0 – Hostile network (zero trust)"]
        DPI[DPI / censor / passive recorder]
        MITM[active on-path attacker]
    end
    subgraph Z1["Zone 1 – Client device (semi-trusted: holds its own keys)"]
        Core["helix-core: WG + transport + kill-switch + DNS"]
        KS["(OS keystore: WG priv + leaf priv)"]
    end
    subgraph Z2["Zone 2 – Gateway VPS (trusted root of control)"]
        direction TB
        Edge["Rust edge (data plane): kernel WG + verdict maps"]
        CP["Go control plane: identity / pki / policy / coordinator"]
        DB["(Postgres: identity/policy/topology + audit)"]
        RDS["(Redis: ephemeral presence + rate-limit)"]
    end
    subgraph Z3["Zone 3 – KMS / HSM (highest trust, smallest surface)"]
        CA["(Tenant CA key – AS-CA)"]
    end
    subgraph Z4["Zone 4 – Private network (trusted by its operator only)"]
        Conn["Connector: outbound-only agent"]
        LAN[LAN hosts behind advertised CIDRs]
    end

    Core -- "TB-1 WG Noise IK (data)" --> Edge
    Core -- "TB-2 mTLS control (leaf cert)" --> CP
    Conn -- "TB-3 mTLS control" --> CP
    Conn -- "TB-3 WG Noise IK (data)" --> Edge
    CP -- "TB-4 sign request (no raw key)" --> CA

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CP -- "TB-5 RLS-scoped SQL" --> DB
 CP -- "TB-6 TTL'd presence" --> RDS
 Edge -- "TB-7 policy verdict ALLOW/DENY" --> LAN
 DPI -. "observes ciphertext only" .- Core
 MITM -. "cannot complete IK / mTLS without keys" .- Edge

Boundary	From → To	Crossing protection	Primary STRIDE risks
TB-1	Client → Edge (data)	WG Noise IK; peer pubkey via policy-filtered map	Spoofing, Information disclosure
TB-2	Client → CP (control)	TLS 1.3 mutual, ≤24 h leaf, pinned CA (S4)	Spoofing, Tampering, Elevation
TB-3	Connector → CP/Edge	same control mTLS + WG IK	Spoofing, Elevation
TB-4	CP → CA	KMS sign grant; CP never holds the raw key (S11)	Elevation, Tampering
TB-5	CP → Postgres	WithTenant under FORCE ROW LEVEL SECURITY (P6)	Elevation (cross-tenant), Disclosure
TB-6	CP → Redis	TTL'd presence; no durable flow data (S6/I5)	Disclosure (privacy)
TB-7	Edge → LAN	nftables/eBPF verdict map; default DROP (I6/R3)	Elevation (lateral), Tampering

4. Attacker classes

Six classes, ordered by capability. Each STRIDE/LINDDUN threat in §5/§6 names the classes that can reach it — a threat reachable only by a nation-state is weighted differently from one reachable by any passive observer.

Class id	Attacker	Capabilities	Cannot (under assumptions §1.3)	Primary goal
C-PASV	Passive network observer / DPI censor	Records all ciphertext on the path; classifies protocols; blocks by fingerprint	Decrypt; inject without detection	Identify VPN use; censor; harvest-now-decrypt-later
C-ACT	Active on-path / MITM	Drop, delay, reorder, inject, replay; spoof endpoints; attempt downgrade	Complete WG IK or mTLS without a valid key (A3/A4)	Hijack a session; downgrade crypto; block recovery

Class id	Attacker	Capabilities	Cannot (under assumptions §1.3)	Primary goal
C-PEER	Malicious authenticated peer	A legitimately-enrolled device that turns hostile; can probe the overlay	Reach peers/CIDRs its compiled policy does not grant (S1/I6)	Lateral movement; reach unauthorized LAN hosts
C-DEV	Compromised client device	Root/admin on a device; reads app memory; drives the OS	Extract a keystore-sealed key without defeating the OS keystore (A2)	Exfiltrate keys; impersonate the user
C-INS	Malicious insider / operator	Admin on the Gateway host; can read Postgres, run code, edit config	Read traffic content/destinations that were never stored (S6); silently sign certs without an audit row (S7) — <i>see residual §10</i>	Surveil users; covertly grant access
C-NAT	Nation-state	Union of C-PASV+C-ACT at scale; legal compulsion of an operator; long-horizon cryptanalysis	Same crypto floor as A3/A4; cannot retroactively read PQ-protected sessions (S10)	Mass surveillance; targeted deanonymization

5. STRIDE per component

Notation: each row is T-<comp>--<X>--<n>, $X \in \{S, T, R, I, D, E\}$. **Mitigation** cites the owning spec; **Residual** is the honest gap (§11.4.6).

5.1 Connector (outbound-only, network-side agent)

The connector dials *out* to the Gateway and advertises CIDRs; it never accepts inbound from the Internet [04_ARCH §1.1]. That outbound-only posture is itself the largest mitigation.

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-CONN-S-1	Spoofing	A rogue agent claims to be a	C-ACT, C-INS	AS-DB	Connector authenticates with a CA-signed device cert (S4); <code>bind_kind=connector</code> on its enroll token; advertised prefixes	A compromised connector <i>can lie</i>

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
		trusted connector and advertises a CIDR it does not serve			are tied to the authenticated device_id [v03-control-plane/svc-policy.md §1.1 ConnPrefixes]	CIDRs enrolled → §10 CONN
T-CONN-T-1	Tampering	Inject/alter advertised-prefix messages on the wire	C-ACT	AS-DB	Advertisement rides the mTLS control channel (TB-3); off-channel injection fails TLS auth	none u A4
T-CONN-R-1	Repudiation	Operator denies advertising a malicious route	C-INS	AS-AUDIT	connector.prefixes.changed is audited with actor binding (S7)	insider DB wr edit au §10 R-
T-CONN-I-1	Info disclosure	Connector leaks the LAN topology behind it	C-PEER, C-INS	AS-MAP	Need-to-know: only devices whose policy grants the connector's CIDR receive it (S3, R2)	a grant device legitim learns slice (design)
T-CONN-D-1	DoS	Flood the Gateway with bogus advertise/reconnect churn	C-PEER	AS-DB	Per-tenant + per-source rate limits on control RPCs (Redis, [04-security-privacy-pki.md §3.4]); recompile is debounced	sustain authen abuse R-DO
T-CONN-E-1	Elevation	Connector tries to be an exit node / full-tunnel target	C-PEER	AS-DB	ERR_EXIT_IS_CONNECTOR — the compiler blocks any exitNodes entry resolving to a connector (P-level, [v03-control-plane/svc-policy.md §5.1 / E7])	none – mecha rejecte compi

5.2 Gateway — Go control plane

The control plane authors identity, signs certs (via KMS), compiles policy, and runs the coordinator. It is the trust root for *who exists* and *who may reach what*.

id	Cat	Threat	Attacker	Asset	Mitigation
T-CP-S-1	Spoofing	Unauthenticated party calls a control RPC	C-ACT	AS-DB	mTLS device cert required on every Coordinator RPC (S4, TB-2); REST behind OIDC/RBAC ([04-security-privacy-pki.md §2.3])
	Spoofing		C-ACT	AS-DB	

id	Cat	Threat	Attacker	Asset	Mitigation
T-CP-S-2		Forged OIDC ID token to gain admin			ID-token signature verified against pinned IdP JWKS; iss/aud pinned ([04-security-privacy-pki.md §2.1])
T-CP-T-1	Tampering	Cross-tenant write via a logic bug	C-INS, C-PEER	AS-DB	WithTenant under FORCE ROW LEVEL SECURITY as non-superuser helix_app (P6, TB-5) — RLS is the backstop <i>below</i> app RBAC
T-CP-T-2	Tampering	Push an allow-all policy by corrupting compile	C-INS	AS-DB	Pure deterministic compiler (P2) + fail-closed dry-run (P4): a bad spec is <i>rejected</i> , never silently widened to allow-all [v03-control-plane/svc-policy.md §5]
T-CP-R-1	Repudiation	Operator denies a privileged change	C-INS	AS-AUDIT	Every state-changing action → audit_events with actor (S7); live stream to Console
T-CP-I-1	Info disclosure	Read another tenant's policy/topology	C-INS, C-PEER	AS-DB, AS-MAP	RLS scope (P6); member role cannot read cross-tenant even with crafted WHERE ([v03-control-plane/svc-policy.md §9])
T-CP-D-1	DoS	Compile-bomb: a huge spec stalls the service	C-PEER (operator role)	AS-DB	Compile is 0(devices × rules × targets) bounded <200 ms at 1k devices (E10/§11 budget [v03-control-plane/svc-policy.md])
T-CP-E-1	Elevation	Member role activates a policy	C-PEER	AS-DB	requireRole("admin", "operator") + RLS floor; agents are read-only w.r.t. policy ([v03-control-plane/svc-policy.md §9])

5.3 Gateway — Rust edge (data plane)

The only component reachable from the hostile Internet on the data port, hence the most hardened (S8). It sees **ciphertext only** and enforces the verdict map.

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-EDGE-S-1	Spoofing	Forge a WG peer to inject into the tunnel	C-ACT	AS-TRAFFIC	WG Noise IK: only a peer holding the registered static private key completes the handshake (TB-1)	none under A3
	Tampering					

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-EDGE-T-1		Container escape to edit verdict rules	C-NAT, C-ACT (post-exploit)	AS-TRAFFIC	Rootless Podman, read-only rootfs, DropCapability=ALL + only CAP_NET_ADMIN/CAP_NET_RAW, default-deny seccomp allowlist, distroless/no-shell (S8, [04-security-privacy-pki.md §5])	a kernel-level 0-day in the allowed syscall surface → §10 R-EDGE
T-EDGE-R-1	Repudiation	n/a — edge keeps no per-flow record (S6)	—	—	by-construction: nothing to repudiate (no traffic log exists)	—
T-EDGE-I-1	Info disclosure	Read user traffic at the edge	C-INS	AS-TRAFFIC	Edge sees ciphertext for the WG core; plaintext exists only transiently in-process during forwarding and is never written to disk (S6/I5)	an insider with live process memory access on a relayed (not E2E) hop <i>can</i> see plaintext in flight → §10 R-RELAY (honest)
T-EDGE-D-1	DoS	Handshake/UDP flood on :443	C-ACT, C-NAT	AS-TRAFFIC	WG's stateless cookie-reply anti-DoS (doc 01); host firewall floor; rate-limit unauthenticated handshakes	volumetric DDoS needs upstream scrubbing (deployment doc 09) → §10 R-DOS
T-EDGE-E-1	Elevation	A granted peer pivots to an <i>un</i> granted CIDR	C-PEER	AS-TRAFFIC, AS-MAP	WG AllowedIPs cryptokey routing drops any packet outside the grant + the edge verdict map default-DROP (I6/R3, TB-7) — two independent layers	none — both layers must fail

5.4 Client device (helix-core)

Semi-trusted: it holds its own private keys but is otherwise least-privileged. The biggest client threat is a fully-compromised device (C-DEV).

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-CLI-S-1	Spoofing	Impersonate the user to the control plane	C-DEV, C-ACT	AS-LEAF	mTLS leaf cert + ≤ 24 h TTL (S4); CSR proof-of-possession at enroll prevents enrolling a key you don't control ([04-security-privacy-pki.md §3.4])	C-DEV that already roots the device → §10 R-DEV
T-CLI-T-1	Tampering	Patch the client to disable the kill-switch	C-DEV	AS-TRAFFIC	Kill-switch is a core-owned state machine that re-enforces the firewall <i>before</i> every transition (S9, [04-security-privacy-pki.md §8.4]); signed app + platform integrity	a rooted device can patch anything — out of the device's own TCB (honest)
T-CLI-R-1	Repudiation	n/a at the client (audit is server-side)	—	—	control writes the client triggers are audited server-side (S7)	—
T-CLI-I-1	Info disclosure	Extract WG/leaf private keys from the app	C-DEV	AS-WGK, AS-LEAF	Keys sealed in OS keystore (Keychain/ Secure Enclave, Android StrongBox,	a device whose keystore is already defeated (A2 false) →

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-CLI-D-1	DoS	Drain battery / wedge the tunnel	C-DEV	—	TPM, DPAPI); Zeroizing on drop; FFI exposes <i>operations</i> not raw bytes (S2, [04–security–privacy–pki.md §3.3]) local-only impact; auto-reconnect ladder + Blocked lockdown (S9) prevents a <i>leak</i> during churn	§10 R-DEV device-local; not a system threat
T-CLI-E-1	Elevation	App reads beyond its sandbox	C-DEV	AS-WGK	Platform sandbox + least-privilege tunnel entitlement (doc 06 shims)	OS sandbox escape — out of scope §1.2

5.5 Coordinator (WatchNetworkMap fan-out)

The coordinator streams policy-filtered maps and revocation deltas. Its central security job is **need-to-know filtering before the wire** (S3).

id	Cat	Threat	Attacker	Asset	Mitigation
T-COORD-S-1	Spoofing	Open a WatchNetworkMap stream as another device	C-ACT, C-DEV	AS-MAP	mTLS device cert binds the stream to a device_id (S4); the server filters <i>for that device id</i> , not a client-supplied one
T-COORD-T-1	Tampering	Forge a map delta to grant a peer	C-ACT	AS-MAP	Deltas originate server-side from the compiled CompiledPolicy; the stream is server→client over authenticated mTLS

id	Cat	Threat	Attacker	Asset	Mitigation
T-COORD-I-1	Info disclosure	A device learns peers it cannot reach (over-share)	C-PEER, C-DEV	AS-MAP	Server filters to <code>CompiledPolicy.VisibleTo[sel before serialization (S3/R2, [v03-control-plane/svc-policy.md § ARTIFACT 1])</code> ; revoked devices never enter visible (P1)
T-COORD-D-1	DoS	Many streams exhaust fan-out resources	C-PEER	AS-MAP	Bounded streams; the <1 s convergence budget is measured under load ([v03-control-plane/svc-policy.md §11])
T-COORD-E-1	Elevation	A device pushes a spec via the RPC	C-PEER	AS-DB	Coordinator RPC is read-only w.r. policy — devices receive maps, never author them ([v03-control-plane/svc-policy.md §9])

5.6 PKI / tenant CA

The CA is the highest-value asset (AS-CA). Compromise mints arbitrary trusted identities.

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-PKI-S-1	Spoofing	Get the CA to sign a cert for a key the requester doesn't hold	C-PEER	AS-LEAF	CSR proof-of-possession verified before signing ([04-security-privacy-pki.md §3.4 / §4.7])	none
T-PKI-T-1	Tampering	Issue a long-lived or wildcard cert	C-INS	AS-CA	Cert profile fixes ≤ 24 h validity + <code>CA:FALSE</code> + <code>clientAuth</code> only ([04-security-privacy-pki.md §4.3]); profile enforced in the signer, not a request field	insider editing the profile code → audited, code-reviewed (§11.4.142)
T-PKI-R-1	Repudiation	Covert cert issuance with no record	C-INS	AS-AUDIT	<code>device.enrolled</code> / <code>device.cert.issued</code> audited (S7); KMS produces its own sign-event log (A1)	insider with both DB <i>and</i> KMS log access → §10 R-CA (highest residual)
T-PKI-I-1	Info disclosure	Read the CA private key	C-INS, C-NAT	AS-CA	Key in KMS/HSM, never in process memory; <code>pki</code> holds a <i>signing grant</i> not the raw key (S11, A1,	a KMS-credential theft → §10 R-CA

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-PKI-D-1	DoS	Flood SignDeviceCert to exhaust KMS quota	C-PEER	AS-CA	[04-security-privacy-pki.md §4.7 CASigner]) Enroll rate limits (§3.4); renewal rides the <i>existing</i> authenticated channel (no new token)	quota exhaustion under authenticated abuse → §10 R-DOS
T-PKI-E-1	Elevation	A revoked device keeps control access	C-DEV	AS-DB	Revocation in <1 s (S5): cert serial blacklisted, open stream force-closed, WG peer removed; ≤24 h expiry is the defence-in-depth floor ([04-security-privacy-pki.md §4.6])	the sub-second race window is honest, not zero (§10 R-RACE)

5.7 Data plane (the tunnel on the wire)

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
T-DP-S-1	Spoofing	Impersonate the gateway endpoint	C-ACT	AS-TRAFFIC	WG IK binds to the gateway's registered static pubkey delivered in the (authenticated) map; a wrong key fails the handshake	none under A3
T-DP-T-1	Tampering	Modify ciphertext in flight	C-ACT	AS-TRAFFIC	ChaCha20-Poly1305 AEAD: any tamper fails the auth tag (doc 01)	none under A3
T-DP-I-1	Info disclosure (now)	Read tunnel content today	C-PASV, C-NAT	AS-TRAFFIC	WG AEAD encryption; no plaintext on the wire	none classically (A3)
T-DP-I-2	Info disclosure (future)	Harvest-now-decrypt-later once	C-NAT	AS-TRAFFIC, AS-PSK	Hybrid PQ PSK (ML-KEM-768) mixed into the WG	PQ is capability-negotiated ; a peer without PQ

id	Cat	Threat	Attacker	Asset	Mitigation	Residual
		quantum matures			handshake — attacker must break <i>both</i> classical ECDH and the PQ KEM (S10, [04–security–privacy–pki.md §9])	falls back to classical (still secure <i>today</i>) → §10 R-PQ
T-DP-D-1	DoS	Block/throttle the tunnel; censor the protocol	C-PASV, C-ACT, C-NAT	AS-TRAFFIC	Transport obfuscation / pluggable carriers + a reconnection ladder (doc 01); kill-switch keeps the leak window closed during recovery (S9)	a total network block is unbeatable from inside — honest (§10 R-CENSOR)
T-DP-I-3	Info disclosure (metadata)	Traffic-analysis: infer activity from packet timing/size	C-PASV, C-NAT	AS-TRAFFIC	Encryption hides content; obfuscation hides the fingerprint; timing/volume correlation is not fully defeated	a global passive adversary can do traffic analysis → §10 R-TA (honest, hard)

6. LINDDUN privacy threats

LINDDUN asks the questions a privacy VPN must answer. Most mitigations here are the no-logging-by-construction family (S6/S7/I5) plus the anonymous-enroll posture.

id	Cat	Privacy threat	Attacker	Asset	Mitigation	Residual
LP-L-1	Linking	Link a user to the sites they visit (the cardinal privacy harm)	C-INS, C-NAT	AS-TRAFFIC	No durable connection/traffic log exists (S6) — there is no dst-bearing row to link from; CI	a <i>live</i> on-box insider can observe in-flight relayed traffic (not stored) → §10 R-RELAY

id	Cat	Privacy threat	Attacker	Asset	Mitigation	Residual
LP-L-2	Linking	Correlate a device's sessions over time	C-INS	AS-PRESENCE	schema-lint fails the build if a flow-shaped table appears ([04-security-privacy-pki.md §6.2]) Presence is ephemeral in Redis (TTL'd); <code>last_seen_at</code> is coarse and carries no destination ([04-security-privacy-pki.md §6.1]) Anonymous mode: enroll token with no email, no SSO;	coarse online/offline timing is observable while live
LP-I-1	Identifying	Tie a tunnel to a real human identity	C-INS, C-NAT	AS-DB	<code>users.email = NULL</code>, <code>oidc_sub = NULL</code>; no reverse-link to a human by design ([04-security-privacy-pki.md §2.2])	managed-mode tenants <i>do</i> store email (the tenant's choice; stated, not hidden)
LP-N-1	Non-repudiation (privacy-adverse)	Prove a specific user visited a specific site (court-compellable record)	C-NAT (legal compulsion)	AS-TRAFFIC	The record does not exist (S6) — the operator <i>cannot</i> produce what was never stored, even under compulsion	the operator <i>can</i> be compelled to start logging <i>future</i> traffic at the relay (warrant-canary class) → §10 R-COMPEL (honest)
LP-D-1	Detecting	Detect that a person uses HelixVPN at all	C-PASV, C-NAT	AS-TRAFFIC	Transport obfuscation makes the protocol harder	obfuscation raises cost, does not guarantee undetectability

id	Cat	Privacy threat	Attacker	Asset	Mitigation	Residual
LP-DD-1	Data disclosure	A breach dumps the membership/destination graph	C-INS, C-NAT	AS-DB, AS-TRAFFIC	to fingerprint (doc 01) Destinations are never stored (S6); the DB holds identity+policy, scoped by RLS; CA key in KMS (S11)	→ §10 R-CENSOR a DB breach still discloses the <i>membership</i> graph (who is enrolled, what CIDs exist) — minimized, not zero → §10 R-DB
LP-U-1	Unawareness	User doesn't know what is/isn't collected	—	AS-PRESENCE	The "what is / is not stored" table is a published spec contract ([04-security-privacy-pki.md §6.1]); the no-log promise is mechanically enforced (S6) and auditable	transparency is a doc property; an end user must trust the published build (mitigated by open self-host)
LP-NC-1	Non-compliance	Operator drifts into unlawful/over-broad logging	C-INS	AS-TRAFFIC	S6 schema-lint fails the build if a connection-log table is added; S7 limits audit to control actions only	an operator running a <i>forked</i> build can re-add logging (self-host honesty) → §10 R-FORK

7. Threat → mitigation map

Every mechanism in [04-security-privacy-pki.md] is justified by the threats it closes; every threat above is closed by a cited mechanism. The forward map (mechanism → threats it defends):

Mechanism (owner)	Invariant	Threats it defends
WG Noise IK + ChaCha20-Poly1305 (doc 01)	—	T-EDGE-S-1, T-DP-S-1, T-DP-T-1, T-DP-I-1
Two-channel auth (mTLS control / WG data), never conflated	S4	T-CP-S-1, T-CLI-S-1, T-COORD-S-1, T-PKI-E-1
	S2	T-CLI-I-1, T-PKI-S-1

Mechanism (owner)	Invariant	Threats it defends
Device keys never leave device; OS-keystore-sealed		
Need-to-know map filtering before the wire	S3 / R2	T-CONN-I-1, T-COORD-I-1, LP-L-1(partial)
Short-lived ≤ 24 h leaf certs	S4	T-CLI-S-1, T-PKI-T-1, T-PKI-E-1
Sub-second revocation pipeline	S5	T-PKI-E-1, T-COORD-* (stale grant)
No durable connection/traffic log + CI schema-lint	S6 / I5	T-EDGE-R-1, LP-L-1, LP-N-1, LP-DD-1, LP-NC-1
Control-action audit (actor-bound)	S7	T-CONN-R-1, T-CP-R-1, T-PKI-R-1
Edge hardening (rootless Podman, seccomp, no-SSH)	S8	T-EDGE-T-1, T-EDGE-D-1
Kill-switch + DNS-leak state machine	S9	T-CLI-T-1, T-DP-D-1 (leak window)
Hybrid PQ PSK (never PQ-only)	S10	T-DP-I-2
CA key in KMS/HSM; cattle data-plane nodes	S11	T-PKI-I-1, T-PKI-T-1
Default-deny policy compiler (pure, fail-closed)	S1 / I6 / P1–P8	T-CONN-E-1, T-CP-T-2, T-EDGE-E-1, T-COORD-I-1
RLS FORCE ROW LEVEL SECURITY backstop	P6	T-CP-T-1, T-CP-I-1, T-CP-E-1
Anonymous enroll (no email/ SSO)	—	LP-I-1, LP-N-1

8. Assets × threats × mitigation matrix

The compact cross-reference: for each asset, the dominant threats and the binding mitigation invariant(s). "✓ Phase-2" marks a mitigation that strengthens but is not MVP.

Asset \ concern	Spoof	Tamper	Disclose	DoS	Elevate	Privacy (LINDDUN)	Binding invariants
AS-CA (CA key)	T-PKI-S-1	T-PKI-T-1	T-PKI-I-1	T-PKI-D-1	—	—	S11, A1, ✓ MVP two-tier issuing CA (offline root + online intermediate) Phase-2 per-region/multi-intermediate + HSM root ceremony

Asset \ concern	Spoof	Tamper	Disclose	DoS	Elevate	Privacy (LINDDUN)	Binding invariants
AS-DB (control truth)	T-CP-S-1/2	T-CP-T-1/2	T-CP-I-1	T-CP-D-1	T-CP-E-1	LP-DD-1	P6 (RLS), S1, S7
AS-WGK (device WG key)	—	—	T-CLI-I-1	—	T-CLI-E-1	—	S2, A2
AS-LEAF (leaf key)	T-CLI-S-1	—	T-CLI-I-1	—	T-PKI-E-1	—	S2, S4
AS-MAP (filtered map)	T-COORD-S-1	T-COORD-T-1	T-COORD-I-1, T-CONN-I-1	T-COORD-D-1	—	LP-L-1	S3, R2
AS-TOKEN (enroll token)	T-PKI-S-1	—	—	T-PKI-D-1	—	LP-I-1	Argon2id hash, single/bounded-use TTL
AS-TRAFFIC (user data)	T-DP-S-1	T-DP-T-1	T-DP-I-1/2/3, T-EDGE-I-1	T-DP-D-1, T-EDGE-D-1	T-EDGE-E-1	LP-L-1, LP-N-1	S6, S9, S10, I6
AS-AUDIT (audit log)	—	T-CP-R-1 (forge)	—	—	—	—	S7, ✓ Phase-2 hash-chain
AS-PRESENCE (Redis)	—	—	LP-L-2	—	—	LP-L-2	I5 (ephemeral/ TTL)
AS-PSK (PQ PSK)	—	—	T-DP-I-2	—	—	—	S10 (ephemeral)

9. Attack trees (two worked)

9.1 Goal: read a target user's traffic content

flowchart TD

```

G["GOAL: read user U's plaintext traffic"]
G --> A1["A. Break the tunnel crypto"]
G --> A2["B. Steal U's device WG key"]
G --> A3["C. Read it at the relay edge"]
G --> A4["D. Compel the operator to log"]

```

A1 --> A1a["Break Curve25519 + ChaCha20-Poly1305 (classical) – infeasible (A3)"]

A1 --> A1b["Harvest-now-decrypt-later (quantum) – blocked by

```

hybrid PQ PSK if PQ negotiated (S10)"]
  A2 --> A2a["Defeat the OS keystore / Secure Enclave (A2) – high
cost (C-DEV)"]
  A2 --> A2b["Phish the enroll token – single-use + TTL + Argon2id
(AS-TOKEN)"]
  A3 --> A3a["Live process-memory access on a relayed hop (C-INS)
– RESIDUAL R-RELAY (honest)"]
  A3 --> A3b["Container escape on the edge –
rootless+seccomp+distroless (S8)"]
  A4 --> A4a["Legal compulsion to start FUTURE logging – RESIDUAL
R-COMPEL (no PAST record exists, S6)"]

```

The tree shows the only *practical* leaves are residuals R-RELAY and R-COMPEL — both honestly stated in §10, both about *live* traffic the operator's box necessarily handles, neither a *retrospective* breach (S6 makes the past unrecoverable).

9.2 Goal: reach an unauthorized LAN host (lateral movement)

```

flowchart TD
  G2["GOAL: malicious peer reaches CIDR it was not granted"]
  G2 --> B1["A. Get policy to grant it"]
  G2 --> B2["B. Bypass enforcement at the edge"]
  B1 --> B1a["Author a wider spec – needs admin/operator role (T-
CP-E-1 blocks member)"]
  B1 --> B1b["Corrupt the compiler to allow-all – fail-closed dry-
run rejects (P4, T-CP-T-2)"]
  B2 --> B2a["Send packets outside AllowedIPs – WG cryptokey
routing DROPS (I6)"]
  B2 --> B2b["Slip past the verdict map – default verdict is DROP
(R3, T-EDGE-E-1)"]
  B2 --> B2c["Exploit stale policy after revoke – sub-second
pipeline + fail-static floor (S5)"]

```

Every leaf terminates at a mechanical block — the two independent data-plane layers (AllowedIPs *and* the verdict map) mean both must fail, and the control-plane path is gated by RBAC + the fail-closed compiler.

10. Residual risks — honestly stated

Per §11.4.6, "mitigated" is not "eliminated". These are the gaps that survive the MVP mitigations, each with its honest boundary and the planned/possible strengthening.

id	Residual risk	Why it survives	Strengthening
R-CA	Tenant CA compromise mints arbitrary identities	A single critical secret; KMS theft or a malicious insider with KMS+DB access is catastrophic	MVP already ships the two-tier issuing CA (offline root + online issuing intermediate) so the root stays offline and a compromised issuing

id	Residual risk	Why it survives	Strengthening
			CA is rotated without touching the root, shrinking blast radius ([04-security-privacy-pki.md §4.2], [../v03-control-plane/svc-pki.md §2]); Phase-2 adds an HSM root ceremony with quorum/audited sign + per-region issuers
R-RELAY	An on-box insider can observe <i>live</i> relayed plaintext	A relay edge necessarily decrypts the WG core to forward; nothing is stored (S6) but in-flight memory is readable by root	True end-to-end (peer-to-peer) paths where the gateway only relays ciphertext; documented as a property, not yet universal — UNVERIFIED for all topologies
R-COMPEL	Operator can be legally compelled to start <i>future</i> logging	S6 removes the <i>past</i> record; it cannot prevent a court order to instrument the live path going forward	Warrant-canary-class transparency; open self-host so the user can run their own gateway
R-INS	Malicious insider with DB-superuser can edit/delete <code>audit_events</code> and bypass RLS	RLS is a non-superuser backstop; a superuser is above it (A1 scopes the CA, not the whole DB)	Phase-2 hash-chained audit (<code>meta.prev_hash</code> , [04-security-privacy-pki.md §7]); external append-only audit sink; least-privilege DB roles
R-DEV	A fully-compromised device (C-DEV) can patch the client and use keystore-sealed keys <i>in place</i>	Keys are sealed against <i>extraction</i> , but a rooted device can invoke key operations	Platform attestation (Phase-2, [04-security-privacy-pki.md §3.5]) to require hardware-bound keys
R-RACE	Sub-second revocation/convergence race window	The system targets p99 <1 s, not a globally-atomic instant; the floor is fail-static, never fail-open ([v03-control-plane/svc-policy.md §10.3])	Measured under load (§11 soak); ≤24 h cert expiry is the hard ceiling
R-EDGE	Kernel 0-day in the allowed syscall surface	Seccomp narrows but cannot eliminate kernel attack surface	Phase-2 SELinux type enforcement; signed/repro images (S8 Phase-3)

id	Residual risk	Why it survives	Strengthening
	escapes the edge container		
R-DB	A DB breach discloses the <i>membership</i> graph (who is enrolled, what CIDs)	Destinations are never stored (S6) but identity/policy must be	Minimize stored fields; anonymous mode removes the human link (LP-I-1)
R-PQ	A non-PQ peer falls back to classical (harvest-now-decrypt-later possible)	PQ is capability-negotiated, on-by-default only when <i>both</i> ends support it (S10)	Make PQ mandatory per-tenant policy (Phase-2 toggle); conservative McEliece hybrid
R-TA	Global passive traffic-analysis (timing/volume correlation)	Encryption + obfuscation hide content+fingerprint, not timing	Padding/cover-traffic research (out of MVP scope) — UNVERIFIED as effective
R-CENSOR	A determined network can block the protocol entirely	No inside-the-tunnel mitigation beats a full network block	Pluggable transports / domain-fronting research (doc 01 carriers)
R-DOS	Authenticated abuse (compile-bomb, enroll/handshake flood)	Rate limits bound it; a large legitimate tenant is the measured ceiling	Upstream DDoS scrubbing (deployment doc 09); adaptive rate limits
R-FORK	A self-host operator can run a <i>forked</i> build that re-adds logging	S6 is a build property of the <i>published</i> build; forks are out of our control	Reproducible signed builds (S8 Phase-3) let users verify the binary matches the no-log source

Honest summary (§11.4.6): the MVP closes every *retrospective* confidentiality and every *lateral-movement* threat with mechanical, two-layer enforcement. The surviving residuals are (a) the single critical secret (R-CA), (b) *live* on-box observation/compulsion at a relay (R-RELAY/R-COMPEL), (c) the insider-vs-audit gap (R-INS), and (d) the unbeatable-from-inside network adversary (R-TA/R-CENSOR). None is a hidden bluff; each has a stated Phase-2 path or an explicit "out of scope" boundary.

11. Test & validation mapping (§11.4.169)

A threat model is itself anti-bluff: each high-severity threat **MUST** have a test whose PASS is captured evidence the mitigation is *live*, not promised ([v03-control-plane/svc-policy.md §12 pattern], §11.4.5/§11.4.69/§11.4.107/§1.1).

Threat(s)	Test type	Concrete test point	Captured evidence
T-EDGE-E-1, T-CONN-E-1, B2a/B2b	E2E / full-automation (§11.4.98)	enroll client; reach an <i>authorized</i> LAN host; deny an <i>unauthorized</i> one — packet-level	docs/qa/<run-id>/transcript + pcap of the DROP
T-CP-T-2, LP-NC-1	Integration + meta-test (§1.1)	inject a CREATE TABLE flows(src,dst,bytes,ts) migration → S6 schema-lint MUST FAIL ([04-security-privacy-pki.md §6.2])	lint FAIL log + restore PASS
T-PKI-E-1, R-RACE	Integration	device.revoke → assert WG peer gone + stream force-closed + serial blacklisted, timed <1 s p99	revoke-to-removed timer trace
T-CLI-T-1, T-DP-D-1	Runtime (§11.4.107)	drop tunnel mid-transfer; capture host physical iface; assert zero plaintext (incl. :53) egresses during Reconnecting/Blocked (S9)	host-interface pcap
T-CLI-I-1, T-PKI-S-1	Security/unit	attempt to read a sealed key via FFI → blocked; CSR with mismatched key → sign refused	refusal capture
T-CP-I-1, T-CP-E-1	RLS test (P6)	tenant-A session cannot read tenant-B rows even with crafted WHERE; member role cannot activate policy	denied-query capture
T-DP-I-2	Integration	negotiate PQ; assert the WG handshake carries the ML-KEM-derived PSK; non-PQ peer falls back classically (S10)	handshake capture
T-EDGE-T-1	Security/chaos	attempt execve/mount/ptrace inside the edge container → seccomp EPERM	seccomp denial log
T-COORD-I-1	Integration	a device whose policy grants only conn-A receives a map with only conn-A; conn-B absent (S3)	served-map byte-compare

Anti-bluff floor: a security "PASS" is valid only with captured evidence that the *attack was actually attempted and blocked* — a green unit assertion that the mitigation "exists" is necessary, never sufficient ([04-security-privacy-pki.md §6.2 self-validated-analyzer pattern]). The §1.1 mutations (strip the schema-lint; widen the verdict default to ACCEPT; drop the revoked-serial check) are the runtime signatures (§11.4.108) that S1/S5/S6/I6 are live.

Sources verified

- [04-security-privacy-pki.md] HelixVPN Volume-4/security spine — §0.1 invariants S1–S11, §1 trust model + two channels, §2 identity (OIDC +

- anonymous enroll), §3 enrollment + S2 key handling + §3.4 anti-replay + §3.5 attestation, §4 PKI/cert lifecycle/revocation <1 s, §5 edge hardening (rootless Podman/seccomp/no-SSH), §6 no-logging-as-code + CI schema-lint, §7 audit, §8 kill-switch/DNS-leak state machine, §9 hybrid PQ PSK. (Read 2026-06-25.)
- [v03-control-plane/svc-policy.md] — §0.1 invariants P1–P8, §3 CompiledPolicy {Version, SpecHash, VisibleTo, AllowedIPs, Verdicts, Via6, ExitNodes}, §5 fail-closed dry-run, §9 authz/RLS, §10.3 honest convergence-race boundary, §12 test matrix. (Read 2026-06-25.)
 - [v02-data-plane/routing-and-addressing.md] — §0.2 invariants I5/I6/R1–R4 (no-logging, default-deny, converge <1 s, need-to-know, fail-closed verdict map). (Read 2026-06-25.)
 - [04_ARCH] HelixVPN-Architecture-Refined.md §1.1 (roles), §3.4 (overlay/policy), §7 (zero-trust, no-logging, edge hardening), §10 (CA/secrets), §12 (Phase-3 repro builds) — cited via the spine doc.
 - [04_P1] HelixVPN-Phase1-MVP.md §6 (identity/PKI/enroll), §7 (policy), §11 DoD (kill-switch, authorized/denied acceptance) — cited via the spine doc.
 - [04_P2] Phase-2 PQ notes §5 (PQ PSK, ML-KEM, McEliece, Rosenpass) — cited via the spine doc.
 - [research-pki_pq_nat], [SYNTHESIS §1/§3/§4/§7] — trust gradient, D4 4via6, key decisions.
 - STRIDE (Microsoft threat-modeling) and LINDDUN (privacy threat-modeling, KU Leuven) are the two methodology frameworks applied; their category sets are used as defined. **UNVERIFIED** items (measured revocation/convergence p99 under 10k streams; universality of the ciphertext-only-relay property; traffic-analysis padding effectiveness) are flagged inline per §11.4.6 and tracked as residuals (§10), never asserted as results.